Motor Selection & Torque Estimate

1. Robot mass

The current estimated robot mass is approximately 4.0 kg, based on the component estimates below.

Component	Estimate or Measured	Mass
Chassis	Measured	2.00 kg
Battery	Measured	0.35 kg
Livox Avia LiDAR	Measured	0.50 kg
Motor drivers	Measured	0.10 kg
Wiring and power distribution (estimate)	Estimate	0.30 kg
Snowplow assembly	Estimate	0.75 kg
TOTAL		4.00 kg

2. Estimated snow resistance

The robot has a nominal width of 13.5in, assuming the plow is marginally wider than the drive base. This would give an approximate plow width of 14in. With an estimate of clearing 1in of snow, it gives a snow cross-sectional area of:

$$A = (14 \times 0.0254) \times (1 \times 0.0254) = 0.00903\ m^2$$

Since the project is still in its early design stage, the actual force required to push snow is not known. For an initial estimate, assume the snow resistance force is 25% of the robot's weight. This is a simple placeholder assumption, based on knowledge of driving in the snow.

$$\text{Robot weight} = m \times g = 4.0 \times 9.81 = 39.2\ N$$

$$\text{Estimated plowing force} = 0.25 \times 39.2 = 9.8\ N$$

A reasonable preliminary range to compare against is 10% to 50% of robot weight (3.9–19.6 N). This range is included to show how sensitive the motor requirement is to the unknown snow resistance.

3. Estimated motor torque requirement

Assume four motors share the load equally and use a preliminary drivetrain efficiency of 70%. The tires are 4.75 in in diameter, giving a wheel radius of 2.375 in (0.0603 m). This is the radius used in the torque estimate.

$$\text{Torque per motor} = (\text{plowing force} \times \text{radius}) \div (\#\ \text{of motors} \times \text{efficiency})$$

$$T = (9.8\ N \times 0.050\ m) \div (4 \times 0.70) = 0.211\ N{\cdot}m \text{ per motor}$$

$$0.211\ N{\cdot}m \approx 2.15\ kgf{\cdot}cm \text{ per motor}$$

Assumed resistance	Plowing force	Required torque per motor*
10% of robot weight	3.9 N	0.86 kgf·cm
25% of robot weight (baseline)	9.8 N	2.15 kgf·cm
50% of robot weight	19.6 N	4.30 kgf·cm

4. Motor comparison

From the RobotShop website, 2 potential candidates were chosen based on price, availability and mounting system.

Specification	Installed GHM-16\nRB-Hsi-05	Candidate RB-DFR-444	Candidate RB-DFR-679
Voltage	12 V	12 V	12 V
No-load speed	200 rpm	251 rpm	350 rpm
Continuous / operating torque	0.78 kgf·cm continuous	3.60 kgf·cm at peak output	2.90 kgf·cm at max-efficiency point
Maximum / stall torque	5.98 kgf·cm	18 kgf·cm	12 kgf·cm
Stall current	1.6 A	7.0 A	5.5 A
Encoder	Not included (installed afterwards)	Hall encoder	Hall encoder
Motor mass	120 g	205 g	98 g

5. What the comparison suggests

Motor	Torque information compared with baseline estimate
GHM-16 (current)	0.78 kgf·cm continuous — as expected this is below the required force for the robot
RB-DFR-444	3.60 kgf·cm at peak efficiency, which is above the expected 2.15 as found in calculations.
RB-DFR-679	2.9 kgf·cm at the maximum-efficiency point; 5.8 kgf·cm at the maximum-power point. While stronger than the GHM-16, it is still below the calculated threshold.

Based on this preliminary estimate, the installed GHM-16 motors appear unlikely to provide enough continuous torque for the assumed plowing load. While the RB-DFR-444 are the strongest, that comes at a cost of weight and require significantly more power. For this reason, the team has decided to go with the RB-DFR-679 motors which are closer in form factor to the default motors and are strong enough to move the robot.

Product sources

- Lynxmotion GHM-16 (RB-Hsi-05): https://ca.robotshop.com/products/lynxmotion-12vdc-200rpm-078kg-cm-ghm-16-w-rear-shaft
- RB-DFR-444: https://ca.robotshop.com/products/12v-dc-motor-251rpm-encoder
- RB-DFR-679: https://ca.robotshop.com/products/12v-dc-motor-12kg-cm-350rpm-encoder

Note: This is an early-stage estimate. The assumed snow resistance, wheel radius and drivetrain efficiency are not measured values. Verify product specifications before procurement.